Conjectures Related to *Lights Out*

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1 Definitions

Let *Lights Out* be a game played on a simple graph $G = (V, E)$ where clicking a vertex v toggles the on/off state of v and its neighbors. One wins the game by finding a sequence of clicks that turns off all the lights.

Definition 1. *The Most Clicks Problem (MCP) asks, for playing on some graph G, what is the most clicks needed to solve any solvable starting configuration, assuming that one solves each configuration in as few clicks as possible.*

That is, what is the most number of moves one needs to be able to solve any solvable initial configuration? This question is akin to looking for "God's Number" on a Rubik's Cube.

Definition 2. *Let $d(n) = \dim(\ker(A + I))$ over the field $\mathbb{Z}_2$ where A is the adjacency matrix of G an $n \times n$ grid.*

You may also hear $d(n)$ referred to as the *nullity* of the $n \times n$ grid. The right-hand side of the equation applies for any graph, not just grids. Elements of the null space correspond to sets of clicks that have no net effect on the states of the vertices when applied. These elements also correspond exactly to subsets of V such that all vertices have an even number of elements from the subset in their closed neighborhood (themselves and their neighbors). Thus, these elements of the null space may be referred to as *null patterns*, *quiet patterns*, or *even dominating sets*.

Definition 3. *Let $F_n(x)$ be the degree n polynomial in the ring $\mathbb{Z}_2[x]$ such that*

$$F_n(x) = \begin{cases} 0 & n = 0 \\ 1 & n = 1 \\ xf_{n-1}(x) + f_{n-2}(x) & n > 1 \end{cases}.$$

These polynomials become useful in calculating $d(n)$. It is known that $d(n) = \deg\gcd(F_{n+1}(x), F_{n+1}(x+1))$.

2 Conjectures

We'd like to extend our existing formulas for $d(n)$, which we have attacked so far by looking at the prime factorization of $n+1$ or the factorization of F_n. Although we list many below, we suspect that are many more similar relations with prime factorization to be found.

Conjecture 1. *For all $n, m, o \in \mathbb{N}$, the following hold:*

$$\begin{aligned}
d(3^n * 5^m - 1) &= 4 \\
d(3^n * 7^m - 1) &= 24 \textit{ for } n > 1, m > 0, \textit{ else } 0 \\
d(2^n * 3^m * 5^o - 1) &= 3 * 2^{n+1} - 2 \\
d(3^n * 11^m - 1) &= 20 \\
d(5^n * 7^m - 1) &= 4 \\
d(3^n * 13^m - 1) &= 0 \\
d(2^n * 3^m * 7^o - 1) &= 13 * 2^{n+1} - 2 \textit{ for } m > 1, o > 0, \textit{ else } 2^{n+1} - 2.
\end{aligned}$$

Notice that these formulas mostly don't depend on the exponents of the primes themselves, except for n on 2.

We can take certain parts of Conjecture 1 and state them more generally.

Conjecture 2. *Let a be an odd number that is not divisible by 21. Then for all $n \in \mathbb{N}$,*

$$d(a^k - 1) = d(a - 1).$$

We suspect that this conjecture is likely false, or at least, not completely descriptive of what's going on. A similar version of this conjecture with $k \geq k_0$ and $d(a^k - 1) = d(a^{k_0} - 1)$ seems to hold for all odd numbers, and it seems we only need $k_0 = 2$. We're not at all sure what's so special about the number 21. We're also not entirely sure that a being odd is what truly matters, especially considering 21 is such an unexplained exception. Instead, we might be able to extend what a's are applicable and deal with the exception of 21 by seeing when d outputs 0 for exponents of 1 and when δ_n is 0 or 2.

We know that $d(n) = 26$ is the smallest even impossible value for d. We also know that if a is an impossible even value of d that $2a + 2$ is also impossible. We would like to better classify and understand impossible values.

Conjecture 3. *Let I be the set of even values that are never values of $d(n)$. Let a family of impossible values be from the recurrence $a_{n+1} = 2a_n + 2$. Then the following are true*

1. *$26 \in I$ (already known)*
2. *M is infinite (already known)*
3. *$34 \in I$ (implies $70, 142, 186, \ldots \in I$)*
4. *$52 \in I$ (implies $106, 214, 430, \ldots \in I$)*
5. *$68 \in I$ (implies $138, 278, 558, \ldots \in I$)*
6. *If $a \in I$, then $2a \in I$ (implies some of the above results)*
7. *I has infinitely many "seed" values. That is, those that are not implied to be in I because a smaller value is also in I.*

Note that the $2a \in I$ part implies that approximately 1 in every 16 even values is impossible. We can see this by taking any even number and dividing it by 2 (e.g. 26 to 13). We start with 13 because we know 26 is impossible. Then with the $2a + 2$ and $2a$ rules we can generate any number that has 13 (i.e. 1101) as a prefix in its binary representation.

Our method for solving the MCP for nullity 2 grids of size $6k - 1$ involved getting an exact solution to a integer linear program by noticing all constraints could be tightly satisfied at once. This allowed us to give a parabola in terms of k that solved the MCP. However, this does not seem to work for higher nullities. We suspect that the answer to the MCP is still a parabola though. Thus, we can calculate the answer the the MCP for 3 boards of the same nullity in the same family (i.e. size is same mod some number) and derive the parabola.

Conjecture 4. *Regarding the answer to the MCP for for an $n \times n$ grid,*

- *If $n = 6k - 1$ and $d(n) = 2$, then the answer to the MCP is $26k^2 - 12k + 1$ (already known).*
- *All nullity 2 grids are of size $6k - 1$ (already known).*
- *If $n = 10k - 6$ and $d(n) = 4$, then the answer to the MCP is $17k^2 - 10k$.*
- *All nullity 4 grids are of size $10k - 6$*
- *If $n = 24k - 13$ and $d(n) = 6$, then the answer to the MCP is $88k^2 - 24k + 1$.*
- *All nullity 6 grids are of size $24k - 13$.*
- *If $n = 40k - 31$ and $d(n) = 8$, then the answer to the MCP is $60k^2 - 20k - 3$.*
- *If $n = 34k - 18$ and $d(n) = 8$, then the answer to the MCP is $161k^2 - 34k - 3$.*
- *All nullity 8 grids are either of side $40k - 31$ or $34k - 18$.*
- *If $n = 60k - 31$ and $d(n) = 10$ then the answer to the MCP is $506k^2 - 60k - 3$.*
- *All nullity 10 grids are of size $60k - 31$.*
- *If $n = 340k - 256$ and $d(n) = 12$, then the answer to the MCP is $3761k^2 - 169k - 13$.*
- *All nullity 12 grids are of size $340k - 256$.*
- ...

Notice that there are different "families" associated with each nullity. These families correspond to a certain size of board. Notice that one nullity might have multiple families. It seems that the smallest board in each family appears in OEIS sequence A118142. We suspect that the parabolas giving the answer to the MCP for a given family always have integer coefficients.

We also have a conjecture about the maximum values of $d(n)$.

Conjecture 5.

$$d(n) \leq \frac{4(n+1)}{5},$$

with equality when $n = 5 \cdot 2^k - 1$.

Our results have already known the equality case, but not that some larger proportion can exist. As far as we know, the only other proof on bounds of $d(n)$ known to date is that $d(4) = 4$ is the only case where $d(n) = n$.

We also have conjectures about $d(p^k - 1)$ for p a prime.

Conjecture 6. *For all primes p,*

$$d(p^k - 1) = d(p - 1).$$

We already almost proven this, just not for when p is a Wieferich prime.

Conjecture 7. *$d(p-1) \neq 0$ for prime p if and only if there exists $n < p-1$ such that $2^n \equiv \pm 1 \pmod{p}$.*

This also means that all Fermat or Mersenne primes p have non-zero $d(p - 1)$.